Predicting Future Anonymized Numeric Target Value Across Multiple Time-Series Entities and Forecast Horizons Using Supervised Machine Learning

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***Abstract*—This project predicts future anonymized numeric target values across multiple time-series entities and forecast horizons using supervised machine learning. The dataset contains over 5 million observations with numerical and categorical features, resembling financial return data. Models tested include Linear Regression, Ridge Regression, LightGBM, CatBoost, and XGBoost, evaluated using weighted error metrics with a time-based validation split. Results show minimal performance differences among most models due to weak predictive signal, while XGBoost performed significantly worse due to poor categorical handling. The best approach was training separate LightGBM models for each forecast horizon, highlighting the importance of horizon-specific modeling.**

I. Introduction

Forecasting future values in time-series data is a key problem in data science, especially in finance and economics. This project predicts a continuous target (y_target) across multiple entities and forecast horizons (1, 3, 10, and 25 steps ahead). Each observation represents a unique entity-time combination, and predictions must use only past information.

The problem is challenging due to its multi-entity, multi-horizon structure and the nature of the target variable, which behaves like financial returns—noisy, volatile, and difficult to predict. Because of this, both simple and advanced models were tested.

To ensure realistic evaluation and avoid data leakage, a time-based validation split was used, training on earlier time steps and testing on later ones.

II. Dataset Overview

The dataset contains 5,337,414 training rows and 1,447,107 test rows, with 86 numerical features and several categorical variables (code, sub_code, sub_category, horizon). Each row represents a forecasting instance defined by entity, time (ts_index), and forecast horizon.

The target distribution (Figure 1) is highly concentrated near zero but contains extreme outliers. The histogram shows a sharp peak around zero, while the boxplot reveals heavy tails. This indicates a skewed, heavy-tailed distribution where most values are small but rare extreme values exist, making prediction difficult.

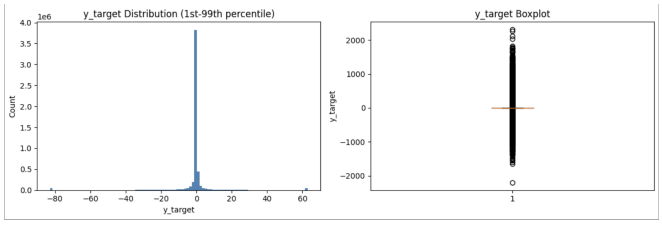

Fig. 1. y target Distribution and Boxplot

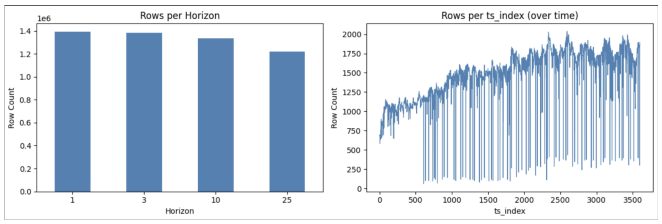

Fig. 2. Rows per Horizon and Rows per ts_index

Dataset structure across horizons and time is shown in Figure 2. The horizons are relatively balanced, though longer horizons have slightly fewer observations. The time-series plot shows variation in row counts over time, including periodic dips, reinforcing the need for a time-based validation approach.

Figures 3 and 4 show how the target behaves over time. The mean fluctuates around zero with occasional spikes, while the standard deviation shows clear volatility clustering. This indicates that the data includes periods of instability, further increasing forecasting difficulty.

III. Models

Multiple models were tested to compare performance. Linear Regression and Ridge Regression served as baselines due

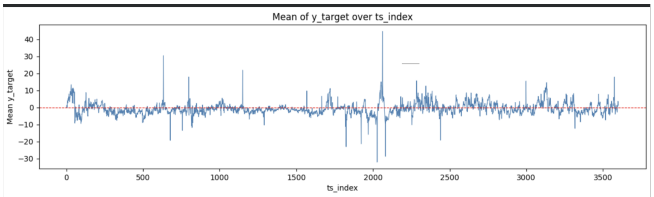

Fig. 3. Mean of y target over ts index

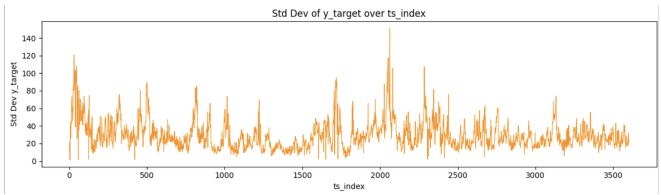

Fig. 4. Standard Deviation of y target over ts index

to their simplicity and efficiency, requiring encoded categorical variables.

Tree-based models included LightGBM, CatBoost, and XGBoost. LightGBM was chosen for scalability, while CatBoost handled categorical features natively. XGBoost required manual encoding, which impacted performance.

In addition to a single global model, separate LightGBM models were trained for each forecast horizon to capture horizon-specific patterns.

IV. Results

Model performance across horizons is shown in Figure 5. XGBoost performed significantly worse than all other models, likely due to ineffective categorical encoding. Linear Regression, Ridge Regression, LightGBM, and CatBoost all performed similarly, with errors near zero. This suggests that the dataset contains limited predictive signal, and complex models do not significantly outperform simpler ones.

Performance worsens as forecast horizon increases, which is expected since longer-term predictions are more uncertain. This supports the use of separate models for each horizon.

Overall, the best approach was per-horizon LightGBM models. While CatBoost performed well as a single model, LightGBM better captured horizon-specific behavior.

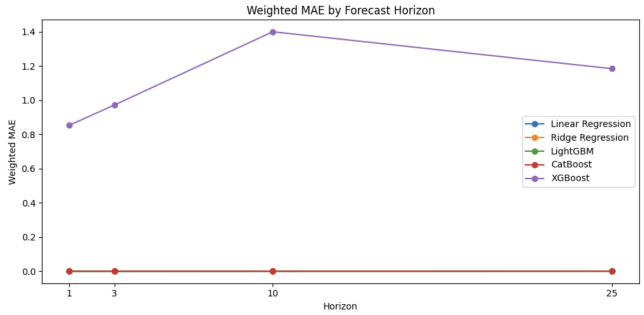

Fig. 5. Weighted MAE by Forecast Horizon

V. Limitations

A key limitation is the target variable itself. Because it behaves like financial returns, it is highly noisy with many near-zero values and rare extreme outliers, making accurate prediction difficult.

Another limitation is the anonymized feature set, which prevents meaningful feature engineering and limits interpretability.

Computational resources also posed challenges. Initial work was done in Google Colab, but even with memory optimizations, the free version did not provide sufficient RAM for later stages. As a result, the project was partially migrated to a Kaggle notebook, which allowed completion of all models but required adapting the workflow across environments.

The dataset size (over 5 million rows) further increased complexity, requiring memory optimization techniques such as downcasting and selective processing.

Finally, model performance depended on preprocessing choices. XGBoost, in particular, struggled due to categorical encoding limitations.

VI. Conclusions

This project highlights the difficulty of forecasting in a large, multi-entity time-series dataset with a noisy target. Most models performed similarly, indicating limited predictive signal.

The best-performing approach was training separate LightGBM models for each forecast horizon, allowing better alignment with the problem structure. Results also showed that longer horizons are harder to predict and that volatility changes over time.

Overall, the project emphasizes the importance of time-based validation, proper handling of categorical variables, and adapting models to the structure of the data. Despite the difficulty of the task, the final approach provides a practical and effective solution.